

APPLICATION OF SHARED VISION PLANNING FOR DROUGHT MITIGATION CLIMATE CHANGE ADAPTATION COUNCIL IN KOREA

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ABSTRACT

Damage due to multi-year droughts mainly caused by climate change are increasing worldwide, and Korea was also affected by the drought that took place from 2015 to 2017. Also, traditional unilateral decision making processes to alleviate the damage from droughts have resulted in conflicts between many involved groups, the need for active participation from both stakeholders and policymakers has become more important than before. In this research, 'Shared Vision Planning,' a collaborative approach that involves participation from various groups of stakeholders, was applied in a research area by organizing a Water Policy Council. Moreover, a 'Shared Vision Model' was developed with the stakeholders by using STELLA Architect software and by applying the concept of System Dynamics. Multiple simulations that include various future climate change scenarios were selected to measure the risk from potential droughts. As a result, the source of water supply, proved to be more vulnerable than the districts in all three aspects of performance indices, reliability, resiliency, and vulnerability. Moreover, although total water deficit in the districts was not very significant, discordance between the districts where the alternatives to prevent water shortage is being developed and where water deficit is projected to happen should be managed. In the future, the model will be further modified according to the stakeholders' needs, with possible seminars to educate the stakeholders for easier use of the developed model.

Keywords: Multi-year Drought, Shared Vision Planning, System Dynamics, Stakeholder Participation Model

1 INTRODUCTION

Frequency of multi-year droughts due to climate change is increasing, and Korea has also been affected by the increased frequency and magnitude of droughts. In order to alleviate the stress on the dam, the government enforced various alternatives including restrictive water rationing for 127 days, and the citizens who rely on Boryeong Dam for municipal and agricultural water supply had to suffer from water shortage. Due to the sudden water shortage, the local government urgently applied various measures to overcome drought, but stakeholders were not included in the decision making process. Lack of discussion between decision makers and stakeholders led to conflicts among various interest groups. Since the need for collaborative decision making process is being more important recently both in Korea and in global society, this research applied Shared Vision Planning, which applies traditional water resources planning process under structured public participation with the aid of integrated computer model, by encouraging the involvement of various interest groups, such as local government, related and specialized organizations, non-governmental organizations and community members. A Shared Vision Model was first established by the model developers by applying the concept of System Dynamics, and it was used and updated during the discussion process between the decision makers and stakeholders to reach an agreement on water supply policy and alternative selection. Ultimately, the developed model was used to analyze the future drought risk of the study basin by using future inflow scenarios that takes the risk of climate change into consideration. Since collaborative decision making process was not widely used in Korea in the past, this research aims to act as a precedent study which applied a modified collaborative decision making approach to include unique characteristics of Korea.

2 METHODOLOGY

2.1 Study Area

The study area of this research is mainly Boryeong Dam and the districts that rely on Boryeong Dam for their water supply, located in central region of Korea, which is illustrated in Figure 1. Moreover, Water Policy Council for Climate Change Adaptation in Chungcheongnam-do, which was established in December, 2016 to provide the stakeholders a chance for discussion, acted as a tool for structured public participation, which is one of the key concepts of Shared Vision Planning.

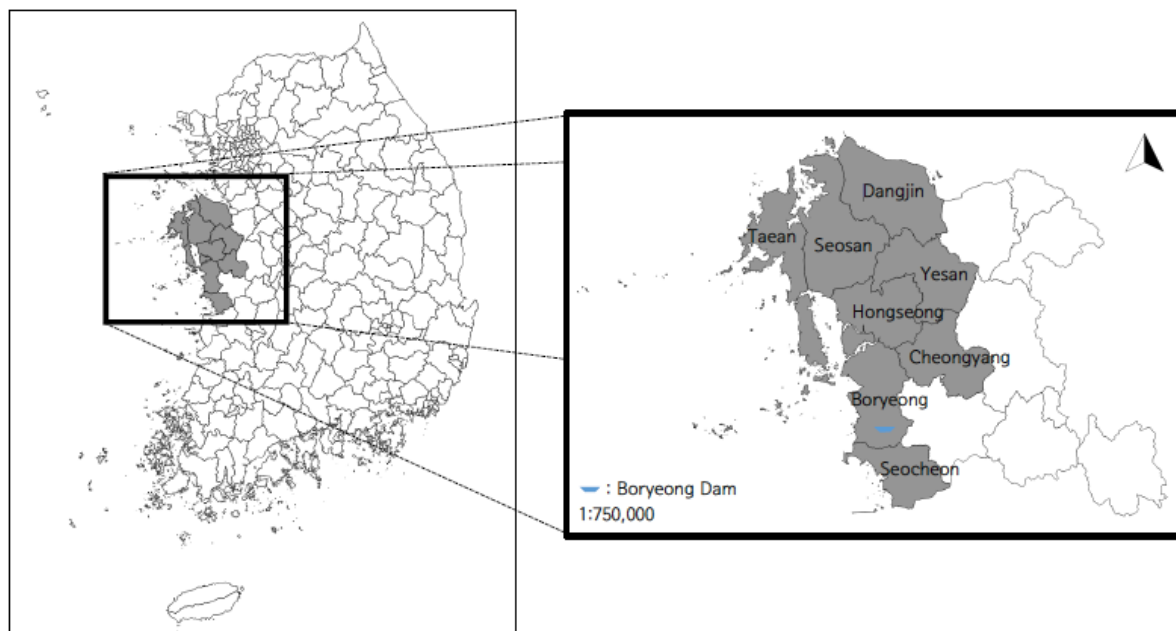


Figure 1. Study Basin: Boryeong Dam and Eight Districts

2.2 Inflow Scenario Selection

In order to analyze the vulnerability of future drought by using the developed model, Global Climate Models can be selected to represent the large range of climate change scenarios. However, due to complexity and computational costs, representative GCMs can be selected to effectively cover the entire range of future climate change. Seo and Kim (2018) purposed a method to select five GCM scenarios among 27 GCMs by using Katsavounidis-Kui-Zhang algorithm which can explain whole-inter-model variability of 27 scenarios and applied it to the national level of South Korea. In order to analyze the risk of future drought, this study selected five representative GCM scenarios in Table 1, which were selected under low flow, Geum-River basin and for 2030s.

Table 1. Selected Representative GCM Scenarios

Scenario Number	Model Name	Resolution	Reference
1	BCC-CSM1-1	2.813×2.791	Wu (2012)
8	CMCC-CMS	1.875×1.865	Davini et al. (2014)
16	HadGEM2-ES	1.875×1.250	Collins et al. (2011)
19	IPSL-CM5A-MR	1.875×1.865	Dufresne et al. (2013)
23	MICROC-ESM-CHEM	2.813×2.791	Watanabe et al. (2011)

3 APPLICATION AND RESULTS

3.1 Development of Shared Vision Model

After the Inaugural Assembly of Water Policy Council in December 2016, total of three council meetings took place until now where the stakeholders from various groups discussed about drought alternatives by using the outcomes of the developed model, which aimed to model the entire water supply system of Boryeong Dam. The model was updated various times with the stakeholders' requests to include certain water resources alternatives unique in the research area. After four council meetings, final Shared Vision Model was established, which was used in the discussion with the stakeholders in the final council meeting. The final model is composed

of 'Water Supply' part illustrated in Figure 2 and the 'Performance Index Calculation' part, which was structured to include the three aspects of frequency, length and magnitude purposed by Hashimoto et al. (1982).

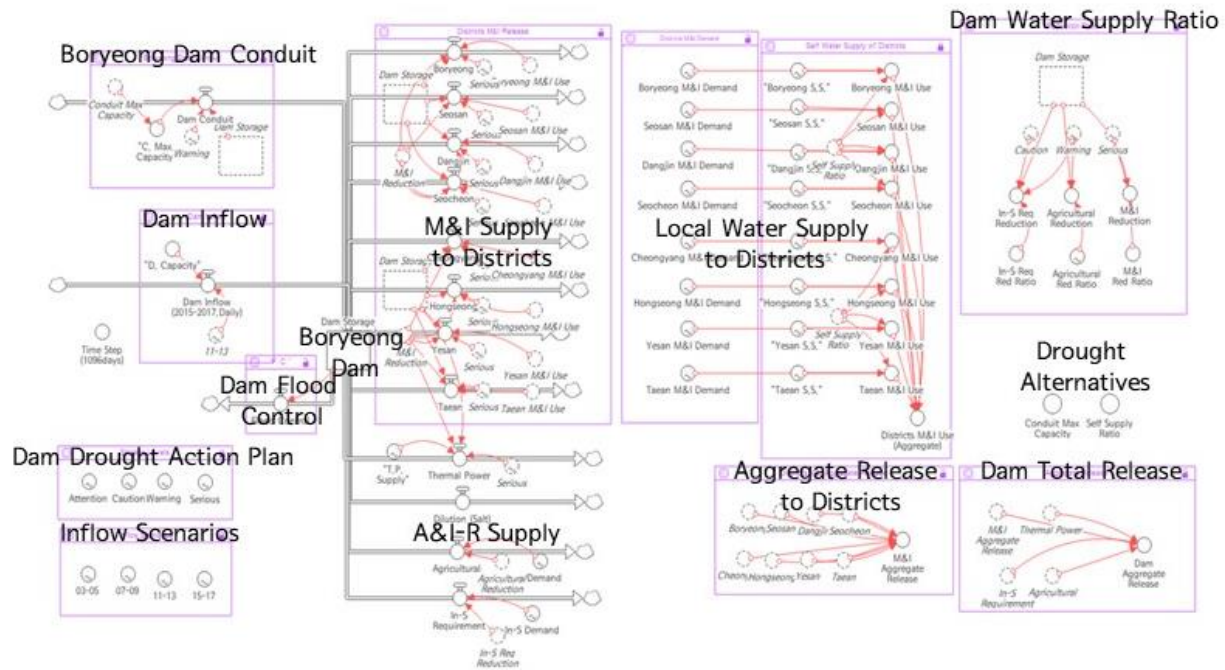


Figure 2. Final Shared Vision Model ('Water Supply' Part)

3.2 Model Configuration

With the final model, four runs as indicated in Table 2 were conducted to analyze the vulnerability of future droughts in the dam and the districts. No drought appeared as the result of the first run, whereas other three runs showed that the dam and the districts may be vulnerable to the risk of future drought. The second run was used for verification of the model since the recorded demand only existed during 2015-2017.

Table 2. Inflow and Demand Scenarios of each Run

Run	Inflow Scenario	Demand Scenario
1	Average Inflow (Observed, 2003-2013)	Actual Water Demand (2015-2017)
2	Drought Inflow (Observed, 2015-2017)	Actual Water Demand (2015-2017)
3	Drought Inflow (Observed, 2015-2017)	Projected Water Demand (2015-2044)
4	GCM Scenarios (Seo and Kim, 2018)	Projected Water Demand (2015-2044)

3.3 Simulation Results

The simulation results were visualized in terms of Alternative Response Functions first suggested by Brown et al., 2012. With the final model, four runs as indicated in Table 2 were conducted to analyze the vulnerability of future droughts in the dam and the districts. No drought appeared as the result of the first run, whereas other three runs showed that the dam and the districts may be vulnerable to the risk of future drought. Among the selected GCM scenarios, it yielded that dam and the districts can be exposed to the risk from future droughts with increased water demand. The simulation results of the worst case GCM scenario of the districts are shown in Figure 2. Purple stars in the figure represent the simulation result from run 3, which represent the potential

risk from future drought with increased water demand as mentioned before. This figure indicates that in the future, districts can be exposed to the drought with similar magnitudes as experienced in the past.

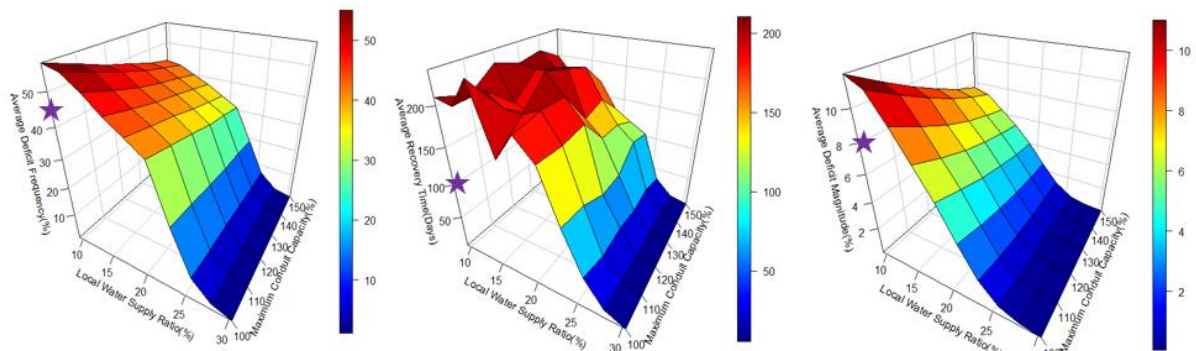


Figure 2. Alternative Response Functions of Districts under Worst Case Scenario

4 CONCLUSION

Key findings and summary of this study can be summarized with followings points:

- Shared Vision Planning was first applied in Korea with a structured stakeholder participation.
- Shared Vision Model was constantly updated with the requests from stakeholders.
- With increasing demand scenarios, both dams and districts can be vulnerable to future droughts under specific climate change scenarios.
- Dams proved to be more vulnerable to future droughts than the eight districts relatively.
- Resiliency, rather than reliability and vulnerability, was harder to predict as shown in the Alternative Response Functions.

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